

5.Data Transformation. Scaling, Binnig .Fixing Skewed Values

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler
df = pd.read_csv("/content/train.csv")
print(df.head())
num_cols = ['Age', 'Fare']
df['Age'].fillna(df['Age'].mean(), inplace=True)
minmax = MinMaxScaler()
df[['Age_MinMax', 'Fare_MinMax']] = minmax.fit_transform(df[num_cols])
standard = StandardScaler()
df[['Age_Std', 'Fare_Std']] = standard.fit_transform(df[num_cols])
print("\nScaling Done:")
print(df[['Age', 'Age_MinMax', 'Age_Std']].head())
df['Age_Group'] = pd.cut(df['Age'],
bins=[0, 12, 19, 35, 60, 100],
labels=['Child', 'Teen', 'Adult', 'Middle', 'Senior'])
print("\nBinning Done:")
print(df[['Age', 'Age_Group']].head())
print("\nSkewness Before:")
print(df['Fare'].skew())    df['Fare_Log'] = np.log1p(df['Fare'])
print("\nSkewness After:")
print(df['Fare_Log'].skew())
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plt.hist(df['Fare'], bins=30)
plt.title("Before Log Transform")
plt.subplot(1,2,2)
plt.hist(df['Fare_Log'], bins=30)
plt.title("After Log Transform")
plt.show()
```

Output:

```
... PassengerId Survived Pclass \
0      1         0        3
1      2         1        1
2      3         1        3
3      4         1        1
4      5         0        3
```

```
0      Braund, Mr. Owen Harris    male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0    1
2      Heikkinen, Miss. Laina    female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0    1
4      Allen, Mr. William Henry    male  35.0    0
```

```
Parch      Ticket      Fare Cabin Embarked
0      0      A/5 21171    7.2500   NaN      S
1      0      PC 17599   71.2833   C85      C
2      0  STON/O2. 3101282    7.9250   NaN      S
3      0      113803   53.1000  C123      S
4      0      373450    8.0500   NaN      S
```

Scaling Done:

```
Age  Age_MinMax  Age_Std
0  22.0    0.271174 -0.592481
1  38.0    0.472229  0.638789
2  26.0    0.321438 -0.284663
3  35.0    0.434531  0.407926
4  35.0    0.434531  0.407926
```

```

Binning Done:
   Age Age_Group
0  22.0     Adult
1  38.0    Middle
2  26.0     Adult
3  35.0     Adult
4  35.0     Adult

```

```

Skewness Before:
4.787316519674893

```

```

Skewness After:
0.3949280095189306
/tmp/ipykernel_3643/2801276784.py:19: FutureWarning: A value is trying to be
The behavior will change in pandas 3.0. This inplace method will never work t

```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.

```
df['Age'].fillna(df['Age'].mean(), inplace=True)
```

